

# SANJEEV KHATIWADA

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## EDUCATION

**Nepal Engineering College (NEC)** | Bachelor of Computer Engineering

Sep 2019 – Aug 2024

- GPA: 3.65/4.00

- Relevant Coursework:** AI, NLP, ANNs, Image Processing & Pattern Recognition, Big Data Technologies

**The New Summit Secondary School** | High School (Science)

Jul 2016 – Aug 2018

- GPA: 3.25/4.00

## SKILLS

**Languages:** Python, C, C++, SQL, NoSQL, Bash and JavaScript

**ML/NLP Tooling:** PyTorch, TensorFlow, NumPy, Pandas, Matplotlib, Scikit-Learn

**Other:** Git, Django, LaTeX, Burp Suite

## PUBLICATIONS & PREPRINTS

**Khatiwada, S.** (2026). *NepOOC-M: Bilingual Nepali-English Benchmark and Comparative Analysis of Multimodal Architectures for OOC Detection*. arXiv:2608.19212. Under review.

## WORK EXPERIENCE

**Things Cyber Pvt. Ltd** | Cybersecurity Intern

Dec 2024 – Nov 2025

- Conducted vulnerability assessments and documented security issues in core banking software systems.
- Audited multi-factor authentication mechanisms, transaction logic workflows, and comprehensive audit trails.

## SELECTED PROJECTS

**NepBiz E-Commerce** | Python, Django, Scikit-Learn, Pandas, NumPy

2024

- Designed a memory-based k-NN collaborative filtering system within a full-stack e-commerce platform; benchmarked against popularity baselines, improving long-tail product recommendation precision by 20%.
- Integrated secure payment gateways (eSewa) and robust user authentication protocols using Django's built-in security features.

**Fleet Management System** | Django, Flutter, REST API

2023

- Architected a comprehensive waste management ecosystem utilizing collection point aggregation and geospatial queries to optimize routes, while integrating real-time notifications and a waste-for-value module to incentivize segregation.

**Self-Driving Agent Simulation** | Godot Game Engine, Python

2021

- Developed a physics-based simulation environment using NEAT algorithm to train autonomous agents for collision avoidance in dynamic driving scenarios.
- Implemented full vehicle dynamics, LIDAR-like sensor simulation with iterative training for trajectory refinement.

## HONORS & AWARDS

**Winner, Research Proposal Competition** | IEEE Pulchowk Campus & CAIR Nepal

Feb 2025

- Co-authored the winning proposal "Analyzing Consumer Behavior in the Digital Age," selected via peer review.

**Academic Merit Scholarship** | Nepal Engineering College

2019 – 2024

- Received a tuition waiver for four Semesters, totaling approximately \$2,500 USD for securing the highest cumulative GPA in the department.

**Programming Award** | Nepal Engineering College

Jan 2021

- Received award for top performance in college-level C++ and C programming competition.

## LEADERSHIP & COMMUNITY ENGAGEMENT

**NAAMII Open House Series**

Aug 2025 – Present

- Engage in monthly sessions to share novel ideas and collaborate with the local research community, gaining insights into AI developments both within Nepal and globally.

**Helambu Education & Livelihood Partnership (HELP Nepal)** | CS Mentor & Educator

Sep 2024

- Mentor over 90 students in Computer Science and Mathematics, empowering them with technical proficiency and fostering a lasting interest in STEM careers.

**Research Writing Workshop** | CAIR Nepal

Feb 2025

- Completed an intensive workshop focused on academic research writing, literature review, and proposal drafting.

**Center for Applied Research and Rural Development (CARRD)** | Executive Member

May 2020 – Aug 2024

- Orchestrated research writing workshops and technical debate competitions, managing end-to-end logistics for student-led academic initiatives.